

Edge Computing Architecture Overview

This document provides an introduction to Edge Computing architecture. It explains how computation moves closer to data sources, enabling reduced latency, improved response time, and efficient IoT workflows.

Topics Covered:

- Edge Processing: Running logic on IoT devices or edge servers
- Fog Computing: Intermediate layer before the cloud
- Device Compute Layers: Sensor → Edge Device → Fog Node → Cloud
- Real-Time Data Handling: MQTT ingestion, low-latency streaming
- Edge Security: Encryption, device identity, secure boot
- Deployment: Docker edge modules, remote updates via Azure IoT Edge

Frameworks Used:

- Azure IoT Edge
- MQTT
- Docker Edge Modules